

Chapter 139

Magic Mask

The Magic Mask palette uses the DaVinci Neural Engine, guided by the user via a click-based interface, to automatically create detailed masks with which to isolate objects or humans, either whole or in part, to which you want to make secondary adjustments in DaVinci Resolve.

This chapter describes how to guide these powerful features using the stroke and tracking controls found within this palette.

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Magic Mask v2 (Studio Version Only)

The underlying AI technology in Magic Mask has been entirely redesigned to both simplify the interface and provide dramatically better results over the previous version and has been designated Magic Mask v2. In addition to better pattern and shape recognition, you will notice the benefits in difficult tracking situations, like objects passing in front of the mask and other depth related issues.

Magic Mask v2 user controls remain mostly the same from the original Magic Mask with the following new changes:

- There are no longer separate Person or Object modes. Magic Mask v2 can handle both on its own.
- Drawing Strokes is no longer necessary and has been replaced by simply performing Clicks on the object to make a selection.
- A simple paint tool has been added to manually paint in or out additional regions from the mask on a frame by frame basis.

The Magic Mask palette uses the DaVinci AI Neural Engine to automatically create a mask to isolate one or more people or objects in the frame, guided by user-applied clicks to identify the subject for isolation. Masks can be generated for either an entire object, person, or for specific features of that person (their face, hair, arms, shoes, etc.). The following images show these two kinds of masks with Highlight enabled in the Viewer. A red onionskin overlay lets you see what Magic Mask is isolating.



(Left) Using Magic Mask to isolate just skin tones, (Right) Using Magic Mask to isolate just the apron. Note how the color of the apron and shirt are very close together, but Magic Mask understands the shape of the apron to generate the mask.

While the masks generated by Magic Mask can often be good enough to use directly for making high-quality isolated adjustments, it won't always give perfect results. In these cases, the Matte Finesse controls let you make the resulting mask softer and looser, as necessary, to clear the edge of a difficult subject you're isolating with the help of another mask generation technique, such as a qualifier key or window. For quick touch up work on a frame, you can also use the paint tool to manually include or exclude regions of the mask.

If you're isolating specific features of a person, you can also mix and match what you're isolating to create exactly the type of mask you require. In the above example, you might isolate the face, along with exposed skin on the torso and arms, to create a mask for creating an overall skin tone adjustment that doesn't include the subject's hair or clothes.

This whole process is guided by clicks you make to identify the subject. In a typical workflow, you'll make positive clicks (colored blue) over the person or feature you want to isolate. Then, if necessary, you can also make subtractive clicks (colored red) over parts of the image that are not the person or feature you're isolating, to correct any problems you see in the generated mask.



Blue clicks identify a person or feature to isolate, while red clicks identify things that shouldn't be included in the mask. In this example we added a positive click (blue) to isolate the wood cutting board, and three subtractive clicks (red) on items that were also roughly wood colored (the wood serving dish and pepper grinder on the left, and her arm) to ensure they would never become part of the mask.

Ideally, both positive and subtractive clicks should be centered on the part of the person or object you're trying to add to the mask. If you don't like the result you're getting in the mask with a particular click you've made, you can drag it to another position without clicking again; clicks are live and can be selected using the pointer to move or delete them.



A single blue click on her back correctly masks her entire body, including arms and hair. Note Magic Mask v2 correctly differentiates her from her reflection in the mirror, even when they are overlapping.

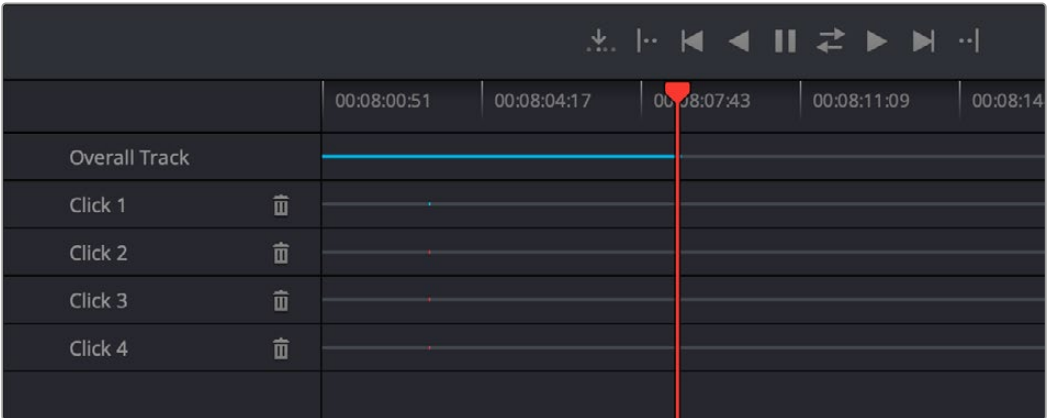
As you work, each click shows up in the Click list, which lets you select, enable/disable, delete, and otherwise manage the different clicks you create to guide automatic mask generation.



The Click list keeps track of all clicks you've made.

Tracking

After you've made one or more clicks to guide the analysis, you can motion track these clicks to follow the subject throughout that shot using the tracking controls within this palette. If you've made multiple clicks, they're all tracked at once, and each click automatically follows along with whatever image details immediately surround that click, so there's no set up necessary. You simply make one or more clicks, then press the track forward or backward button. When you track a click, the overall tracking bar shows you which frames have been tracked (tracked frames are blue).

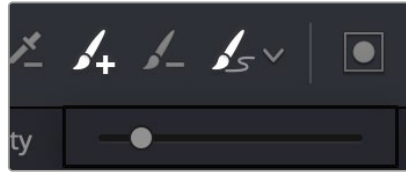


Controls let you track all mask analysis clicks forward and backward through the shot.

Once you've added the clicks that are necessary to isolate the subject, and you've motion tracked them to follow along with the motion of the shot, a mask is automatically generated live for each frame of the shot. This is important to know because any change you make, adding or removing clicks or moving them manually, will alter the resulting mask on the fly.

Fixing Masks with the Paint Tool

Occasionally the Magic Mask might miss its mark over the course of tracking and include something in the mask you don't want, or exclude something that you do. There is a simple paint tool available in the toolbar to patch up holes in the mask, or remove masked areas from the selection. The paint tool works on a frame by frame basis, and so won't track with the mask. It's great for removing short mask hiccups, but for anything more than fixing a few frames, you should go back and refine your mask using the add and subtract tools.



The Add/Subtract Paint Stroke tools and the Brush Size control

The plus and minus paint tools let you add or subtract strokes from the mask as necessary, while the paint brush size tool lets you grow or shrink the brush used to make the strokes.



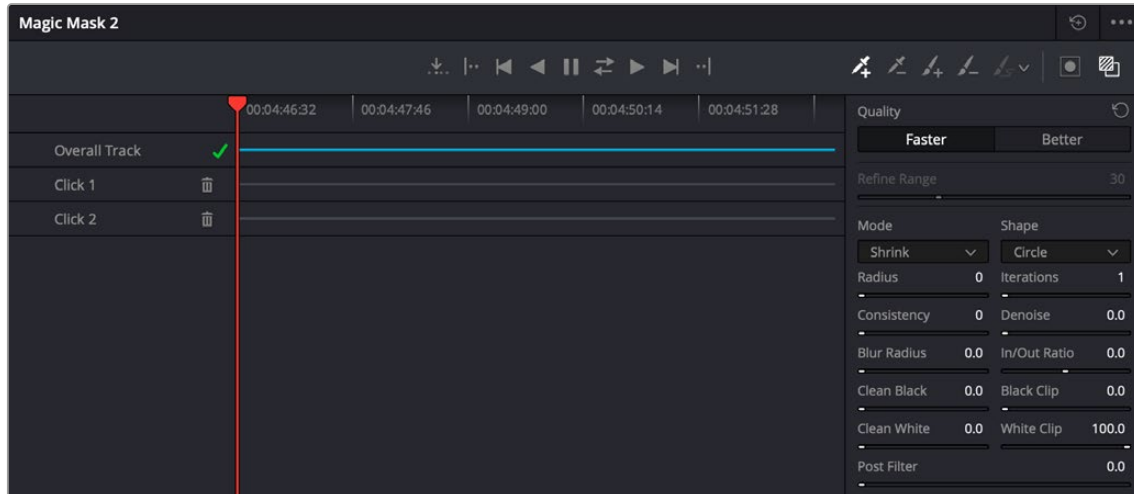
(Left) In this frame, Magic Mask accidentally included the silver rings on her fingers in the mask. For most of the track it did not, so we just need to fix a few frames. (Right) Selecting the Subtract Paint Stroke tool (green circle) and adjusting its size, we carefully paint a hole in the mask over her rings.



The final frame, with the rings correctly excluded from the mask

The Magic Mask Interface

The Magic Mask palette is divided into three sets of controls: the Toolbar, the Click list, and the Mask Finesse panel. When you first open the Magic Mask palette, it's empty, and you're told to click in the Viewer to create a mask.



The controls of the Magic Mask palette

Magic Mask Toolbar

A toolbar at the top contains most of the interactive controls of the Magic Mask palette.

Tracking Controls: The following controls let you track all available clicks to follow camera or subject motion in the frame. From left to right, these include:

- **Go To Reference Frame:** Moves the playhead to the frame on which you initially made the clicks.
- **Go To First Frame of Tracked Area:** Moves the playhead to the first tracked frame of a range of tracked frames in preparation for tracking backwards if there are untracked frames at the beginning of the clip.
- **Track One Frame Reverse:** Tracks one frame backwards and stops. Useful if you're tracking frame by frame to watch the progress of a particularly complicated bit of motion. If something goes wrong, you can back up to the last frame where the click was able to properly track the subject, and drag the click to a better location using the pointer to make it follow the subject properly. If necessary, you can go a frame at a time, dragging the click to a better position every time it fails to follow the feature you're using it to isolate.
- **Track Reverse:** Continuously tracks from the current frame all the way to the beginning of the clip.
- **Stop Tracking:** Stops tracking in cases where there's a problem with the track and you want to make a change.
- **Track Forward and Reverse:** Tracks from the current frame all the way to the end of the clip, then returns to the original tracking point and tracks backwards to the beginning of the clip.
- **Track Forward:** Continuously tracks from the current frame all the way to the end of the clip.
- **Track One Frame Forward:** Tracks one frame forward and stops. Useful if you're tracking frame by frame to watch the progress of a particularly complicated bit of motion. If necessary, you can go a frame at a time, dragging the click to a better position every time it fails to follow the feature you're using it to isolate.
- **Go To Last Frame of Tracked Area:** Moves the playhead to the last tracked frame of a range of tracked frames in preparation for tracking forwards if there are untracked frames at the end of the clip.

Click tools: Two tools at the right let you choose whether to make clicks to identify the feature you want to isolate or identify things that aren't the feature in order to eliminate unwanted excursions in the resulting mask. The plus eyedropper adds an area to the mask; the minus eyedropper removes an area from the mask.

Paint tools: The three paint tools let you do quick touchup work on a mask on a frame by frame basis. The plus paint brush lets you add to the mask, while the minus paint brush lets you subtract from the mask. The paint brush size control lets you change the size of the brush.

Invert Mask: A button lets you invert the resulting mask in cases where you want to use the feature analysis of this palette to isolate everything except the feature or features being analyzed.

Mask Overlay: Turns on an onion-skinned overlay to see what parts of the image are being masked alongside which aren't, so you can continue to refine the result by adding, moving, or deleting clicks. The isolated part of the mask is tinted translucent red.

Click List

Once you start making clicks to identify features for mask generation, they appear in this Click list.

Click list header: The header, at the left of the Click list, has controls for selecting and deleting individual clicks.

Click timeline area: A Timeline ruler shows the duration of the current clip you're creating a mask for. The Overall Track displays how many frames of each click have been tracked. There is a small colored mark (blue or red) on each click track to show where the reference frame is.

Mask Adjustment Controls and Matte Finesse

There are two sets of controls for refining the mask that's output by the Magic Mask palette. The first set of controls, at top, let you adjust how the mask is generated based on the analysis data, which lets you refine the mask result based on characteristics of the image.

Quality: Two options let you choose a tradeoff between quality and performance. Faster lets you generate a lower quality mask more quickly, that's suitable for garbage matting. Better generates a higher quality mask with more detail, that's more processor-intensive.

Smart Refine: In Better mode this control lets you expand or contract the resulting mask based on the analysis of the image, such that expanding the mask doesn't expand it deeply into surrounding parts of the image that aren't connected to that person or object. This operation does not introduce softness. Instead, it enlarges or shrinks the overall mask generated by this palette.

A second set of mask manipulation controls are for manipulating the mask after it's been generated. Most of these are the same Matte Finesse controls that are available in the Qualifier palette, which are useful for trying to fix issues with problem masks or soften the edges when you need to have a more feathered result.

- **Mode/Shape:** Lets you choose how you want to modify the Alpha channel/key. You can choose Shrink or Grow to dilate or erode the edges of the matte with great accuracy. Or, you can choose Opening or Closing to plug or expand holes to clean up a ragged matte. Shape lets you choose the shape of the growth or contraction.

- **Radius:** this slider adjusts how much to shrink, grow, open, or close the edge's key.
- **Smoothing:** Smooths along the temporal frames.
- **Blur Radius:** In small amounts, blurring a key does well to take the edge off problem edges. However, blurring a key can also feather the edges of a key past the border of the subject you're keying, with the result being a visible "halo" around your subject, depending on the adjustment you're making.
- **Clean Black:** Clean Black is a specialized operation that eliminates noise seen as white speckling in the black area of a key. Raising Clean Black lets you "fill holes" in the background portion of a key and erode translucent edges.
- **Clean White:** Clean White is another specialized operation that eliminates noise (seen as black speckling when viewing a high-contrast highlight) in the white portion of a key that includes areas of the image you're isolating and expands the key by making light parts of a key lighter the higher you raise this parameter, pushing light gray areas of the key toward white. The practical result is that raising Clean White lets you "fill holes" in the foreground portion of a key and grow translucent edges.
- **Iterations:** Refines the recognized shape.
- **Denoise:** Provides a distinct way to post-process extracted keys to selectively reduce the noise in a key, getting rid of stray areas of qualification and softly filling holes in a matte.
- **In/Out Ratio:** Using In/Out Ratio can help eliminate fringing. Raising In/Out Ratio will fill in small black holes in the matte, while lowering In/Out Ratio below 0 will eliminate speckling by pushing small white bits of the matte toward black.
- **Black Clip:** Raising Black Clip applies a "lift" adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward black.
- **White Clip:** Lowering White Clip applies a "gain" adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward white.
- **Post Filter:** Performs a final clean up of the key, using the original image for reference; useful for bringing back some fine detail in sharp edges or hair.
- **Consistency:** After you've tracked each click to follow the subject over the duration of the clip, this setting lets you choose how much temporal smoothing is necessary to ameliorate jitter in the edges of the resulting mask in areas of low confidence, such as frizzy hair or translucent clothing. Higher settings apply more smoothing to the edges of the mask but are more processor intensive and may affect how closely the mask follows motion in the image. Lower settings will be faster and more accurate but may allow more edge jitter in the resulting mask, which can be distracting in the final adjustment you're making. This parameter defaults to 0, so your first application of Magic Mask will always begin with the most accurate (and potentially most active) application of this feature's analysis.

IMPORTANT: Consistency requires a click to have a duration of at least a few frames in order to function correctly. This requires you to track each click to follow the motion of the camera and subject in order to extend the duration of each click. Because Consistency is trying to eliminate one or two frame "noise" in the shape of the mask, clicks of short duration may end up having their effect eliminated.

NOTE: The original Magic Mask is still accessible in DaVinci Resolve 20 by selecting Legacy Object Mask from the Magic Mask 2 options menu.

Magic Mask v1 (Legacy Version)

NOTE: The original Magic Mask v1 has been superseded by Magic Mask v2 as described above. However the v1 Magic Mask is still accessible for legacy projects, and described below. To enable it select “Legacy Object Mask” from the Magic Mask 2 options menu.

The Magic Mask palette uses the DaVinci Neural Engine to automatically create a mask to isolate one or more people or object in the frame, guided by user-applied paint strokes to identify the subject for isolation. Masks can be generated for either an entire object, person, or for specific features of that person (their face, hair, arms, shoes, etc.). The following images show these two kinds of masks with Highlight enabled in the Viewer. A red onionskin overlay lets you see what Magic Mask is isolating.

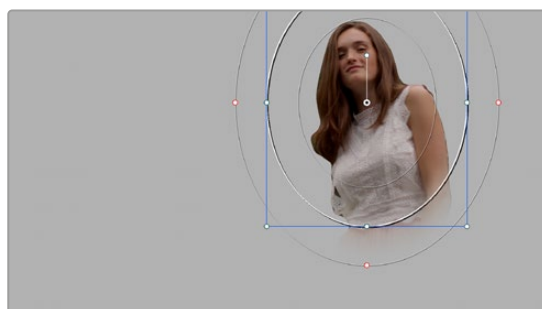
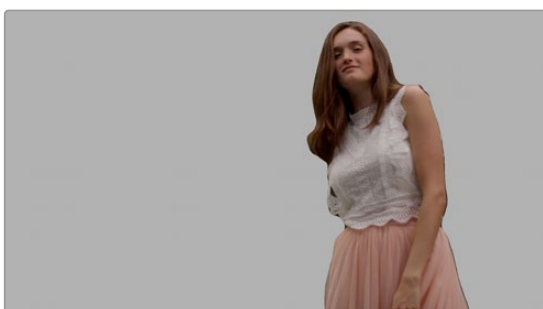


(Left) Using Magic Mask to isolate an entire person, (Right) Using Magic Mask to isolate someone's face

Once the Magic Mask palette has generated a mask, Matte Finesse controls let you adjust the mask's size and feathering. This lets you deal with difficult edges that might tip your hand depending on the kind of adjustment you're trying to make.

While the masks generated by Magic Mask can often be good enough to use directly for making high quality isolated adjustments, it won't always give perfect results. Keep in mind that Magic Mask is as much for the fast creation of detailed garbage masks as it is for doing direct isolation.

In these cases, the Matte Finesse controls let you make the resulting mask softer and looser as necessary to clear the edge of a difficult subject you're isolating with the help of another mask generation technique, such as a Qualifier key or window. When you use the Magic Mask palette within the same node in which you're also creating a key in the Secondary palette, or a window in the Window palette, the key or window and Magic Mask are combined to output a boolean intersection. This means the isolation is limited to areas where both kinds of masks overlap one another.



(Left) Using Magic Mask to create a mask, (Right) Combining the Magic Mask with a circular window, the resulting mask is an intersection of the two

Objects can also be isolated by Magic Mask, allowing you to create complicated masks of complex shapes that may be difficult to isolate with other tools. For example, using the Magic Mask in Object mode to modify only the wood grain section of this guitar.



(Left) Multiple strokes in Object mode isolating the wood grain of the guitar, while ignoring the musician's arm, (Right) The finished shot, with a warmer wood grain

If you're isolating specific features of a person, you can also mix and match what you're isolating to create exactly the type of mask you require. For example, you might isolate the face, along with exposed skin on the torso, arms, and legs, to create a mask for creating an overall skin tone adjustment that doesn't include the subject's hair or clothes.



(Left) Using Magic Mask to isolate a face, (Right) Using Magic Mask to isolate all exposed skin

As described earlier, this whole process is guided by strokes you draw to identify the subject. In a typical workflow, you'll draw short positive strokes (colored blue) over the person or feature you want to isolate. Then, if necessary, you can also paint short subtractive strokes (colored red) over parts of the image that are not the person or feature you're isolating, to correct any problems you see in the generated mask.



Blue strokes identify a person or feature to isolate, while red strokes identify things that shouldn't be included in the mask.

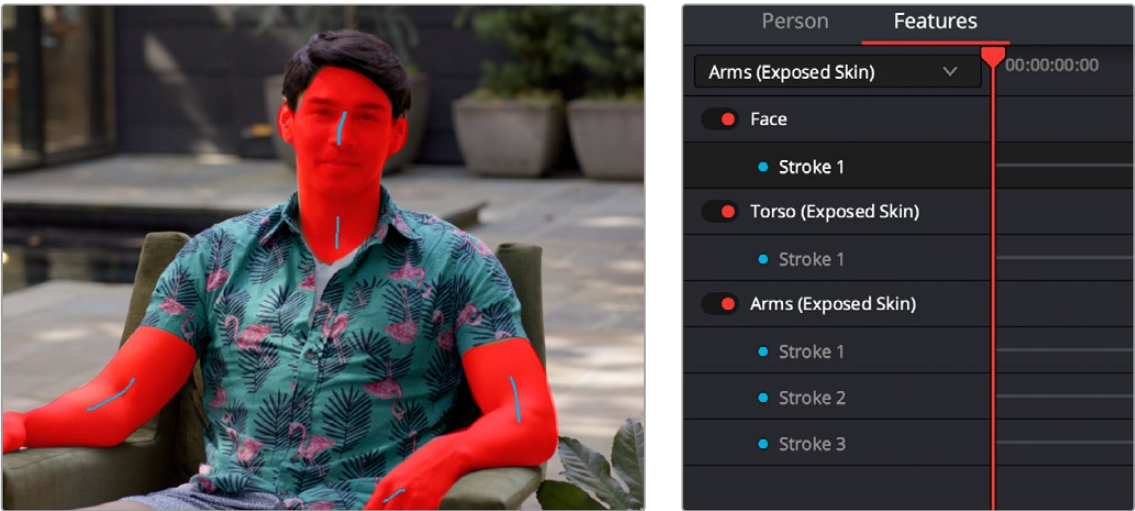
Ideally, both positive and subtractive guidance strokes should be relatively short, and centered on the part of the person or object you're trying to add to the mask. Stroke position is usually more important than stroke length, and in fact if you don't like the result you're getting in the mask with a particular stroke you've drawn, you can drag it to another position without redrawing it; strokes are live and can be selected using the pointer to move or delete them.



Selecting a stroke with the pointer; selected strokes become highlighted.

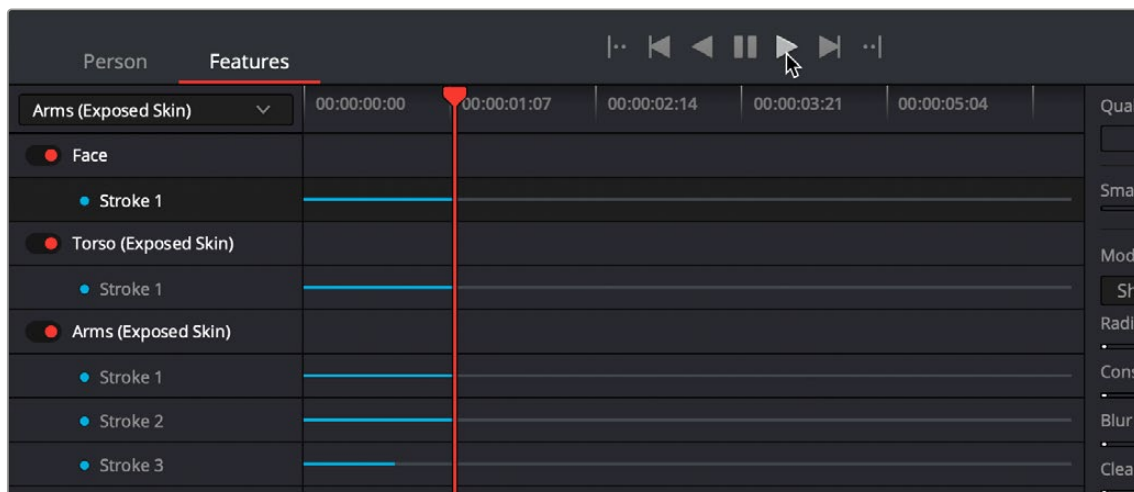
Very long paint strokes aren't usually that useful, and can be counterproductive later on when you're trying to track strokes to follow motion in the clip. This is particularly true if a stroke needs to follow something that's changing shape as it moves, like an arm. Furthermore, while you can make as many strokes as you like, fewer strokes work better than many strokes, and as a general rule you shouldn't use more than five strokes to isolate a particular person or feature (nor should you need to).

As you work, each stroke shows up in the Stroke list, which lets you select, enable/disable, delete, and otherwise manage the different strokes you create to guide automatic mask generation.



(Left) Multiple strokes in Features mode isolating exposed skin, (Right) The Stroke list keeps track of all strokes you've made.

But what about camera and/or subject motion, you ask? After you've painted one or more strokes to guide the analysis, you can motion track these strokes to follow the subject throughout that shot using the stroke tracking controls within this palette. If you've made multiple strokes, they're all tracked at once, and each stroke automatically follows along with whatever image details immediately surround that stroke, so there's no setup necessary. You simply draw one or more strokes, then click the track forward or backward button. When you track a stroke, the tracking bar for that stroke in the Stroke list shows you which frames have been tracked (tracked frames are blue).



Controls let you track all mask analysis strokes forward and backward through the shot.

Once you've added the strokes that are necessary to isolate the subject, and you've motion tracked them to follow along with the motion of the shot, a mask is automatically generated live for each frame of the shot. This is important to know because any change you make, adding or removing strokes or moving them manually, will alter the resulting mask on the fly.

What Magic Mask Is Good For, and What It's Not

Keep in mind that the Magic Mask tool does not specialize in soft translucent mask edges, such as those created by the HSL, Luma, or 3D Keyers (when they're used carefully). On the other hand, it excels at creating form-fitting masks that isolate human features in situations where a typical keyer does poorly. For example, Magic Mask can isolate exposed skin in shots where that person is standing in front of a background with wood, stone, or soil that inhabits the same ranges of color as that person. It can isolate beige clothes against a beige background. It can even isolate a person within a black and white clip.

Once you've drawn one or more strokes to isolate a mask, a variety of controls let you adjust the resulting mask to better fit the subject, if necessary. Depending on the adjustment you're making to the subject at hand, you may find that the keys generated using Magic Mask are sufficient to convincingly isolate the person you're grading, particularly if you're working in Better mode.

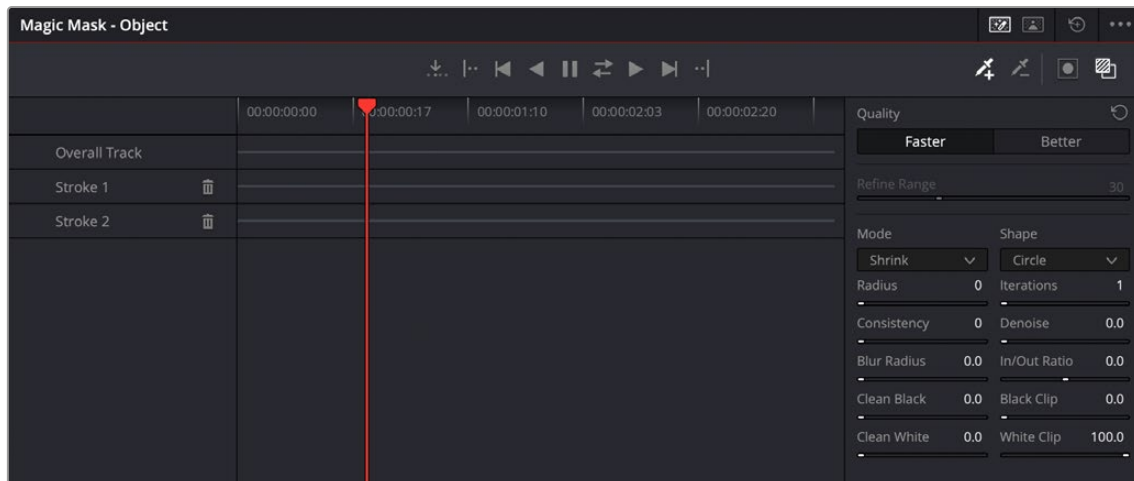
However, real world projects pose plenty of unexpected challenges. Keep in mind that, depending on what you're trying to do, the resulting automated mask isn't always going to be perfect enough to replace precise rotoscoping for intensive secondary color adjustments. However, it's been designed to easily create tight-fitting garbage masks to use in conjunction with secondary keys or windows, with which to exclude the background around a difficult-to-isolate subject.

If you're using Magic Mask to create a garbage mask, keep in mind that the Faster mode has been optimized to create looser masks more quickly for situations where you'll be using it in conjunction with a keyer or window to isolate a specific area for special grading.

TIP: The Magic Mask palette can even isolate people in photos and paintings. However, figurative representations must have a minimal level of shadow rendering and anatomical detail for a figure or feature to be recognized. For this reason, cartoony or abstract representations won't isolate very well.

The Magic Mask Interface

The Magic Mask palette is divided into three sets of controls: the Toolbar, the Stroke list, and the Mask Finesse panel. When you first open the Magic Mask palette, it's empty, and you're told to draw a stroke in the Viewer to create a mask.



The controls of the Magic Mask palette in Object mode

Magic Mask Toolbar

A toolbar at the top contains most of the interactive controls of the Magic Mask palette.

Object Mode: Selecting this icon puts the Magic Mask tool into Object mode for isolating subjects that are not people.

Person Mode: Selecting this icon puts the Magic Mask tool into Person mode for isolating people and parts of people. A tab control at the left lets you choose the type of mask you want created. There are two choices:

- **Person:** For creating a mask to isolate as much of a person is visible, given the framing of the shot.
- **Features:** For creating a mask isolating just someone's face, or hair, or arms, for example. In Features mode, you can also combine multiple specific features into a single mask.

Tracking Controls: The following controls let you track all available strokes to follow camera or subject motion in the frame. From left to right, these include:

- **Go To Reference Frame:** Moves the playhead to the frame on which you initially drew the strokes.
- **Go To First Frame of Tracked Area:** Moves the playhead to the first tracked frame of a range of tracked frames, in preparation for tracking backwards if there are untracked frames at the beginning of the clip.
- **Track One Frame Backwards:** Tracks one frame backwards and stops. Useful if you're tracking frame by frame to watch the progress of a particularly complicated bit of motion. If something goes wrong, you can back up to the last frame where the stroke was able to properly track the subject, and drag the stroke to a better location using the pointer to make it follow the subject properly. If necessary, you can go a frame at a time, dragging the stroke to a better position every time it fails to follow the feature you're using it to isolate.
- **Track All Frames Backwards:** Continuously tracks from the current frame all the way to the beginning of the clip.

- **Stop Tracking:** Stops tracking, in cases where there's a problem with the track and you want to make a change.
- **Track Forwards and Backwards:** Tracks from the current frame all the way to the end of the clip, then returns to the original tracking point and tracks backwards to the beginning of the clip.
- **Track All Frames Forwards:** Continuously tracks from the current frame all the way to the end of the clip.
- **Track One Frame Forwards:** Tracks one frame forward and stops. Useful if you're tracking frame by frame to watch the progress of a particularly complicated bit of motion. If necessary, you can go a frame at a time, dragging the stroke to a better position every time it fails to follow the feature you're using it to isolate.
- **Go To Last Frame of Tracked Area:** Moves the playhead to the last tracked frame of a range of tracked frames in preparation for tracking forwards if there are untracked frames at the end of the clip.

Stroke tools: Two tools at the right let you choose whether to draw strokes to identify the feature you want to isolate or identify things that aren't the feature in order to eliminate unwanted excursions in the resulting mask. The plus eyedropper adds an area to the mask; the minus eyedropper removes an area from the mask.

Invert Mask: A button lets you invert the resulting mask in cases where you want to use the feature analysis of this palette to isolate everything except the feature or features being analyzed.

Mask Overlay: Turns on an onion-skinned overlay with which to see what parts of the image are being masked alongside which aren't, so you can continue to refine the result by adding, moving, or deleting strokes. The isolated part of the mask is tinted translucent red.

Stroke List

Once you start drawing strokes to identify features for mask generation, they appear in this Stroke list.

- **Stroke list header:** The header, at the left of the Stroke list, has controls for selecting, enabling/disabling, and naming each stroke (if you need that level of organization). Right-clicking any stroke in its header reveals controls to Reset Tracking Data and change the mask type (feature or not feature).

In *Object* mode, this is just a flat list of strokes, each of which help to either define the object you want to isolate, or the background.

In *Person* mode, this is just a flat list of strokes, each of which help to either define the person you want to isolate, or the background.

In *Features* mode, an additional drop-down menu lets you choose which specific feature you want to identify with a guidance stroke. All strokes are sorted hierarchically by the feature they correspond to, so for example each Face stroke appears underneath a Face title bar. The title bar for each feature has a toggle control that lets you turn that feature off and on. When off, you disable that feature's contribution to the overall mask being generated.

- **Stroke timeline area:** A Timeline ruler shows the duration of the current clip you're creating a mask for. Each stroke in this list has a track that displays how many frames of each stroke have been tracked. As you track each stroke, these tracks fill up with colored bars to show which frames have tracking data.

Tracks of strokes that identify a person or object are blue, while tracks for strokes identifying things that aren't the feature are red. Using this timeline, you can keep track of which strokes need to be tracked, and manipulate the tracking data as necessary to obtain a useful result.

Mask Adjustment Controls and Matte Finesse

There are two sets of controls for refining the mask that's output by the Magic Mask palette. The first set of controls, at top, let you adjust how the mask is generated based on the analysis data, which lets you refine the mask result based on characteristics of the image. These include:

- **Quality:** Two options let you choose a tradeoff between quality and performance. Faster lets you generate a lower quality mask more quickly, that's suitable for garbage matting. Better generates a higher quality mask with more detail, that's more processor-intensive.
- **Smart Refine (Person Mode):** This control lets you expand or contract the resulting mask based on the analysis of the image, such that expanding the mask doesn't expand it deeply into surrounding parts of the image that aren't connected to that person. This operation does not introduce softness. Instead, it enlarges or shrinks the overall mask generated by this palette. 0.5 is unity, 0 is the maximum amount of expansion, while 1 is the maximum amount of contraction.
- **Refine Range (Object Mode):** This control lets you adjust the sensitivity of the resulting mask based on the analysis of the image. This lets you fine tune the mask to encompass more or less of the selected object. 30 is unity, 100 is the maximum amount of sensitivity, while 1 is the minimum amount of sensitivity.

A second set of mask manipulation controls are for manipulating the mask after it's been generated. Most of these are the same Matte Finesse controls that are available in the Qualifier palette, which are useful for trying to fix issues with problem masks, or soften the edges when you need to have a more feathered result.

For more information about the Matte Finesse controls, see *Chapter 135, "Secondary Qualifiers."*

There is one additional control, however, that's unique to the Magic Mask palette.

Consistency: After you've tracked each stroke to follow the subject over the duration of the clip, this setting lets you choose how much temporal smoothing is necessary to ameliorate jitter in the edges of the resulting mask in areas of low confidence, such as frizzy hair or translucent clothing. Higher settings apply more smoothing to the edges of the mask but are more processor intensive and may affect how closely the mask follows motion in the image. Lower settings will be faster and more accurate but may allow more edge jitter in the resulting mask, which can be distracting in the final adjustment you're making. This parameter defaults to 0, so your first application of Magic Mask will always begin with the most accurate (and potentially most active) application of this feature's analysis.

IMPORTANT: Consistency requires a stroke to have a duration of at least a few frames in order to function correctly. This requires you to track each stroke to follow the motion of the camera and subject in order to extend the duration of each stroke. Because Consistency is trying to eliminate one or two frame "noise" in the shape of the mask, strokes of short duration may end up having their effect eliminated.

Choosing What to Isolate

This section elaborates on the different choices that are available for isolation and offers tips for what each choice is good for.

Object Mode

Object mode is the easiest choice to make. Essentially, use this mode for any subject that is not a person. Subjects like cars, pets, and food are all good candidates for Object mode. The notable exception to this rule are items of clothing like shirts, hats etc. that are being worn by a human. Worn clothing is much better isolated in Person mode instead.

Drawing positive strokes in Object mode will select areas of similar contrast and color, allowing you to link complex shapes together. Generally you will need more strokes to actively define an object than a person, due to the greater variety of the shapes involved.

Drawing negative strokes in Object mode removes areas from the object that you don't want to isolate. This can be something simple like removing the wheels of a car from a mask, or more complicated like removing specific books from a mask of a bookshelf.

Person Mode

When you choose Person, every stroke you draw over someone helps to isolate that person's entire body, or however much of that person is visible onscreen. This includes all clothes that person is wearing, including their shirt, pants, shorts, dress or skirt, bathing suit, knickers, jumpsuits, hat, shoes, even armor or space suit (yes, we've tested both), whatever. Most clothing should be correctly identified as part of the isolation mask you're creating. However, excessively flowing garments may need an extra stroke or two to identify them in complex visual compositions.

Drawing negative strokes, when in Person mode, will identify everything that's not the person you're isolating; this omits both the background and/or other people from the mask being generated.

Drawing negative strokes isn't mandatory; you only need to draw a negative stroke if you see a problem such as part of the background being included in the isolation mask.

Features Mode

When you choose Features, a drop-down menu appears at the top of the Stroke list. You must first choose the feature you want to isolate from this menu, and then draw strokes over the selected feature (face, hair, etc.) you want to isolate. Each feature's collection of strokes can be turned on or off to add that feature to the overall mask being created.

- **Face:** Isolates just the face and ears, including eyes and glasses. This omits hair and the neck (which is considered part of the torso). Facial hair, particularly bigger beards, will usually be omitted from an initial stroke made on exposed skin of the face, so you can work on just the skin if you want. However, an additional stroke on the beard will include it in the face isolation.
- **Hair:** Isolates just the hair, stopping at the bangs and ears, and omitting as much of the face as possible. Curly, frizzy, or wispy hair will produce oversized masks encompassing as much of the hair detail as possible. This tool does not create fine transparency. Fine and irregular hair detail can cause jittery edges, but this can be ameliorated using the Consistency control.
- **Torso (Exposed Skin):** Isolates exposed skin above the waist and below the face, excluding the arms. This is the selection you want to enable if you want to isolate the neck, neckline, and any part of the chest that's exposed when wearing v-necks, plunge necks, tank tops, crop tops, or bikini tops.

- **Clothing (Top):** Any clothing above the waist. Exposed skin is omitted.
- **Clothing (Bottom):** Any clothing below the waist. Exposed skin is omitted.
- **Arms (Exposed Skin):** The exposed parts of arms or hands. Shirt and dress sleeves are omitted, as are sun sleeves.
- **Shoes:** Most styles of shoes will be isolated.
- **Legs (Exposed Skin):** Isolates exposed skin below the torso, excluding socks and leggings.
- **Hat:** Most styles of hats will be isolated, although thin accessories such as streaming ribbons or feathers might not be included.

Remember, no matter what feature you've selected to isolate, you can always draw negative strokes to omit other parts of the person or their clothing that are erroneously being included in the mask.

IMPORTANT: When you're drawing strokes in Features mode, accidentally drawing a stroke over a feature other than the one that's selected in the drop-down menu at the top of the Stroke list will result in a completely incorrect mask. If you see this happen, delete the stroke you just made, double-check which feature is currently selected, and either redo the stroke or choose the correct feature you want to isolate.

Identifying More Than One Person

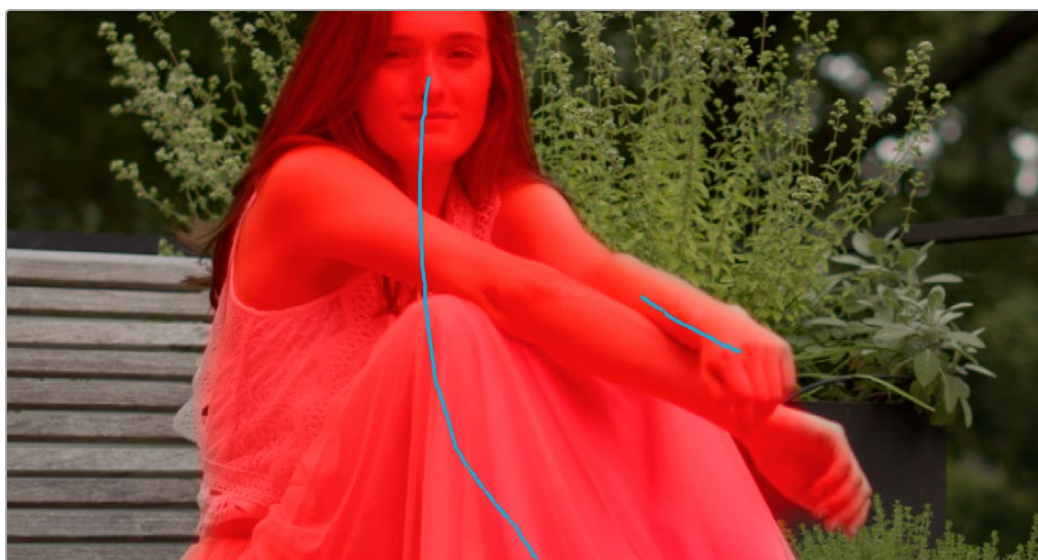
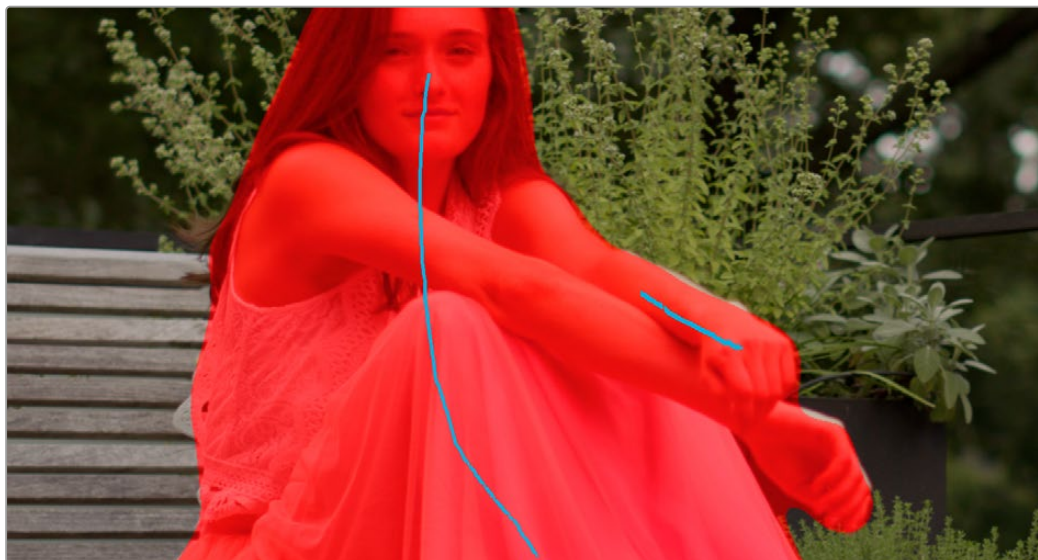
In both Person and Feature mode, you can add strokes to identify more than one person, or more than one person's features. There's nothing special you need to do, simply draw strokes over each person you want to isolate, or over the same feature of each person you want to include (for example, drawing stroke over three people's faces in Face mode).



You can draw strokes over multiple people to isolate them together in the same mask.

Using Matte Finesse Controls

There are plenty of situations where a seemingly problematic mask can be improved using the Smart Refine and Matte Finesse controls at the right of the Magic Mask palette. If the mask is only a few pixels off, and the color adjustment you want to make is reasonably subtle, you can probably use Blur Radius and In/Out Ratio to create a soft edge that falls off just at the outer edge of the subject (so as not to create a halo effect) with which to make your adjustment.



(Top) The hard edges of the initial mask aren't quite close enough to the edges of the woman's arm to make a subtle adjustment, (Bottom) Using the Blur Radius and In/Out Ratio controls to create soft edges that don't extend past the subject to make a useful mask

Using Magic Mask as a Garbage Matte

In situations where the outer edge of a person is not being detected closely enough for Magic Mask to give a useful result for a more extreme color adjustment, you're probably better off expanding the result into a closely-fitting garbage matte to use for isolating the person or features using other techniques, such as a keyer in the Qualifier palette. Even for this purpose, you'll find using

Magic Mask is typically faster than rotoscoping a relatively close garbage matte over a moving figure using windows.



(Top) The initial mask isn't quite close enough to the edges to make an extreme adjustment, (Bottom) Using the Mode/Shape/Radius/Iterations/Blur Radius controls to expand the matte to be a garbage matte for use with other keys

Dealing With Hair

Wispy or frizzy hair, individual strands of hair being thrown back, and tips of dreadlocks or heavily moussed hair that comes to points will often require special handling with this tool. Keep in mind that the Magic Mask tool does not specialize in soft translucent mask edges like a keyer does. When it comes to hair styles that are loose and free, in many situations you may be able to finesse an acceptable result using the Matte Finesse controls to expand a softened version of the matte to encompass the entire hairstyle.

If this interferes too much with the edges of other features of the subject you're isolating, you can create a dedicated hair mask in Features mode by drawing Hair strokes. You can then use the Matte Finesse controls at the right of the palette to create softened edges that work well with the outer boundary of the hair style you're isolating.



Creating a soft-edged hair mask

At this point, you can try doing different color adjustments using two different correction nodes, one for the subject's hair, and another using Magic Mask to isolate everything but the hair.

Alternately, if you want a single matte with which to isolate the entire subject, you can then use additional instances of the Magic Matte tool (one per node), using one node to isolate just the face and body without the hair, and a third node to isolate the entire subject but with a very soft edge. Then, you can use a Key Mixer node to combine all these masks together with the original hair mask you created, with which to feed a complete key to another node used to make the correction.



(Left) Nodes 2, 3, and 4 are creating three different masks that are combined using a key mixer and connected to the Key input of Node 5 to make an adjustment, (Right) The final result of this three-mask approach.

While hair with soft edges can be a challenge, Magic Mask excels at many elaborate hair styles with detailed shapes and hair accessories, including jewelry and ribbons. For example, the stylized hairstyles found in European and Chinese period movies and shows give the DaVinci Neural Engine no trouble at all, particularly as these styles are usually tightly bound with no wispy hair.

Asymmetrical hairstyles may require extra strokes to identify hair that's significantly longer on one side of the face. This may also be true for unorthodox hair styles, such as hair that's been sculpted into distinct sculptural shapes. Lastly, tight braids that expose skin on the scalp are also challenging for the kind of hard masks that Magic Mask produces, so you're better off using Magic Mask as a garbage matte and using a keyer to create a detailed soft mask to use in these instances.

About Hats

Magic Mask works with an incredibly wide variety of hats from all time periods. On hats, tassels or details like feathers can be a problem (in testing, the feather on Errol Flynn's cap in *Robin Hood* couldn't consistently be included, even though the cap itself was no problem). Additionally, veils that are part of hats may or may not be easily identified, but an additional stroke usually solves the problem.

Dealing With Accessories

Things that people are holding, such as handkerchiefs or umbrellas, are best selected in Object mode. Jewelry and watches pose a special problem, as they're typically so thin and small that they can be difficult to omit, but you may not want to include them depending on the adjustment you're trying to make. Your results will vary with the thickness of the items and the framing of your subject, but in general your results will probably be better if you simply add a stroke or two to include these items in the isolation, since omitting fine detail like a necklace can be problematic.

Adding Strokes to Guide Mask Creation

Here are some tips for how to draw strokes to analyze the image. To identify people, try starting with a single short positive stroke (blue), centered on the face or body.



Drawing the first stroke to identify an entire person

Depending on the result, more strokes may be needed to differentiate an arm or frizzy hair from the background. If you're adding a stroke to an arm or hair, don't put it right at the edge; the goal is not to trace the subject, but to use strokes within the body of the subject to clarify what features belong to the subject, and which do not.



Adding a stroke to the arm to include it as well

As you do this, resist the urge to use too many strokes. One stroke might well be all you need, or you might need two or three, but using more than five strokes for any subject or feature may cause more problems than it solves.



Adding negative strokes to the reflection to remove

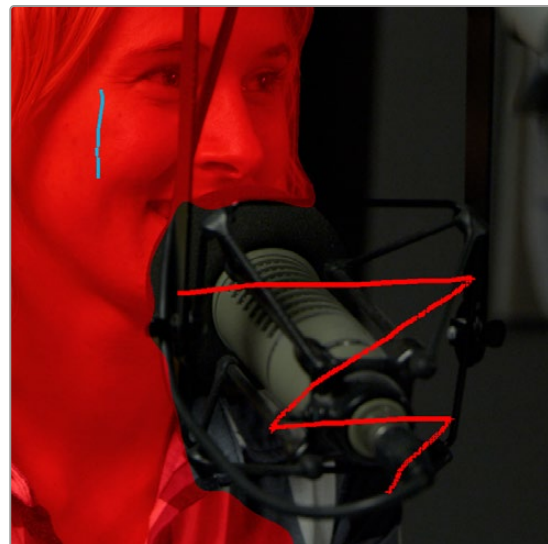
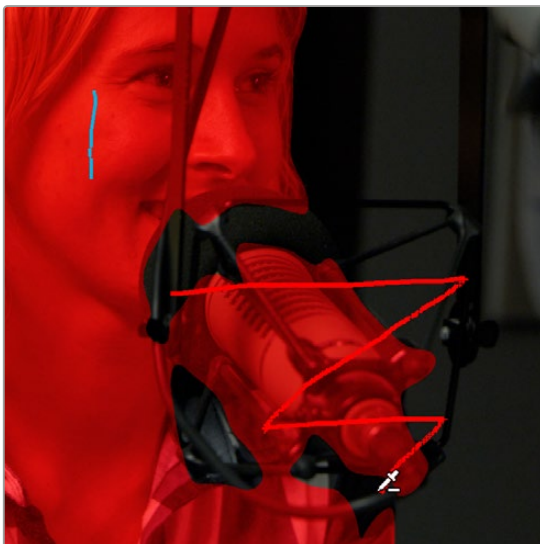
To identify faces, try starting with a single short stroke in the middle of the face, along the nose, or from eyes to the lips. Depending on the result, you may need additional strokes to identify the chin, a beard, or to distinguish the forehead from wispy hair.



Drawing strokes to identify a face; notice that additional strokes are needed to include the beard.

It's a good idea to start with a single stroke, then track the stroke through the duration of the clip (described later in this section). Over the entire clip, you'll see which parts of the mask may exhibit problems. This will guide where you put additional strokes, in order to more clearly identify the person versus the background, in order to clean up the resulting mask. As you add additional strokes, track those as well and see how things go.

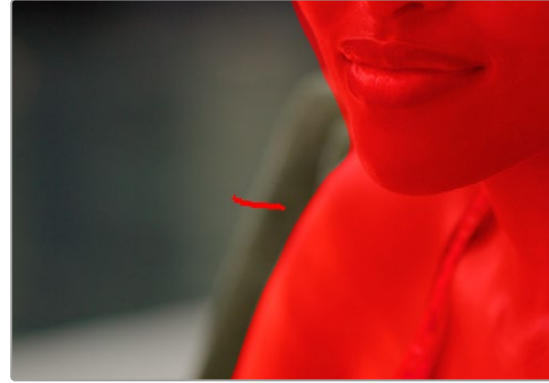
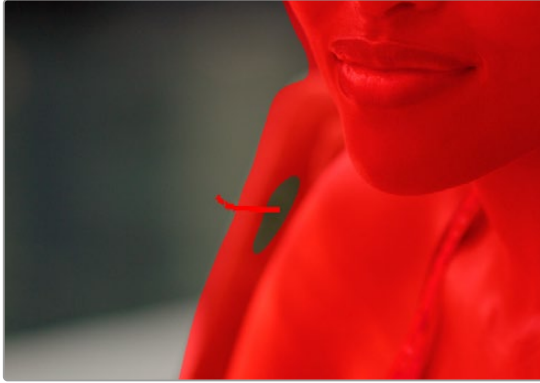
If you notice parts of the background that are erroneously added to the mask, you can add a negative stroke (red) to clarify that those parts of the image should not be isolated. Short strokes, longer strokes, and zig zags are good for backgrounds. Since strokes just help the analysis update the mask, you can try different types of strokes to see what works best, and delete any stroke that doesn't give good results. As with positive strokes, don't use more strokes than are absolutely necessary to get the result you want.



(Left) The microphone incorrectly identified as part of the woman being interviewed, (Right) Drawing negative strokes to identify the microphone as background, using a zig-zag pattern to tag the entire microphone

Similarly, if you're isolating features such as a face, and you find other features, such as hair, getting in the way, you can draw a negative stroke on the hair to identify it as not part of the face.

If there's a "hole" in the resulting mask that's actually part of the background and not the subject, you can try drawing a stroke connecting that hole to the rest of the background.



Drawing strokes to connect the background with itself, to remove part of the mask that's not part of the person

If there's an "island" in the mask that's actually part of the subject and not the background, you can try drawing a stroke across the island that connects one solid part of the mask to itself.



Drawing strokes to fill an island in the subject being isolated

Of course, extracting a mask using strokes is only the first part of creating a useful mask. The Quality, Consistency, and Smart Refine controls let you refine the stroke analysis of the image to create the best tradeoff between performance and quality for the type of mask you need. These controls work hand in hand with the strokes you draw to extract a mask. A set of Mask Finesse controls then let you adjust the resulting mask to manipulate and soften it, to better suit whatever adjustment you're trying to make. All of these controls are described earlier in this section.

Managing Strokes in the Stroke List

When you draw a stroke, it appears in the Stroke list. By default, all strokes last for the duration of the clip to which they're applied, so if you draw a stroke to identify a figure that's not moving, there's no requirement to motion track the stroke. Strokes in Person mode appear in one long list, while strokes in Features mode appear hierarchically, with the strokes for each feature separated by the title of each feature. Once a stroke appears in this list, it has the following controls.

Enable/disable feature: In Features mode, once you draw at least one stroke to identify that feature, the title of that feature appears, underneath which is organized all the strokes for that feature. A toggle next to the feature name lets you turn that feature's contribution to the overall mask on and off.

Enable/disable stroke: Each individual stroke in this list has a blue dot that lets you turn that stroke's contribution to the overall mask on or off.

Stroke Timeline Area with Stroke Tracks: The stroke track for each stroke lets you keep track of and manipulate the animation of strokes to follow camera and subject motion. Blue or red frames in this track let you see at a glance how much of each stroke has been motion tracked; blue frames for positive strokes, red frames for negative strokes. You can also use these tracks to identify frames where you'd like to mute that stroke's contribution to the final mask and to keep track of which frames have been muted. You can draw a bounding box within the stroke track around however many frames of however many strokes for which you want to delete motion tracking.

Stroke Duration

When you first draw a stroke, that stroke has a duration of one frame, located at the timecode position of the playhead when you drew it. If you move the playhead to the left or right, you'll see the stroke disappear. If you want one or more strokes to continue analyzing the subject for the duration of that shot, you need to use the Motion Tracking controls to track the stroke to follow along with the motion of whatever it's drawn on top of. As a stroke is tracked, its duration increases to cover the entire range of tracked frames. If at any point a stroke outlasts its usefulness, such as when a person walks off frame, you can stop tracking that stroke when it's no longer necessary and it will stop contributing to the analysis.

This is useful, because often you'll want to place strokes to deal with analysis issues that only last for a few frames, to fix a hole in the mask that only appears briefly. Just be aware, you can't simply draw a stroke and move on to the next shot; you must track at least one stroke to last for the duration of a subject's time on screen for that subject to be continuously masked.

Tracking Strokes to Follow Subject Motion

You can think of each stroke you've drawn as a persistent eyedropper that samples the image that overlaps it. The mask that results from all strokes' collective analysis of the image is generated live over each frame of the clip. This means that if the camera or the subject moves, you need to motion track or otherwise adjust the position of each stroke to follow along with the motion, so the subject continues to be correctly identified. You also need to make sure that each stroke is able to analyze what it's supposed to, and turn off strokes that can't for whatever reason.

TIP: Magic Mask is a processor-intensive operation, so to accommodate users of less powerful workstations, there is a Use Fast Tracking option available in the Magic Mask palette's Option menu, which speeds up the process of tracking at the expense of potentially less accurate tracking for fast or erratically moving subjects.

Methods of Moving Strokes to Follow Subject Motion

For these reasons and more, there are multiple methods of manipulating strokes to refine the final analysis.

- **Motion Tracking:** A set of tracking controls at the top of the Magic Mask palette let you motion track one or more strokes to follow the camera and subject motion in a clip. Every frame that's motion tracked effectively has a motion-tracked reference frame placed on it that moves the stroke to a new position.
- **Reference Frame:** Any time you drag a stroke from its original position in the Timeline, you create a static reference frame at that frame that instantly moves the stroke from its previous position to the new position at that frame. Because these are similar to static keyframes, there's no motion interpolation from one "keyframe" to the next in order to create animation. Instead, a reference frame in the Magic Mask palette is a simple frame-by-frame operation. Every time you move a stroke at a new frame, you change the stroke's position at that frame from that point forward, without changing stroke position on any other frame that's been previously tracked.
- If this seems simplistic, keep in mind that the goal is not to create beautiful stroke animation with this tool. You only need to move each stroke to follow along with whichever subject or background feature it's supposed to be sampling in each frame. The only important thing is that, in every frame, the stroke overlaps a suitable part of the feature you're isolating to create the appropriate mask. If the stroke itself jumps around abruptly, that doesn't matter so long as the mask that's generated is good. This means that even if you're stuck doing a limited amount of frame-by-frame animation to make a stroke follow along with a troublesome feature (such as a veil flapping in the wind), it'll go pretty fast, because in each frame you're just dragging the stroke to overlap the feature wherever it happens to have moved. This repositioning doesn't have to be smooth, just accurate.
- **Combining Tracking and Reference Frames:** Since each frame that's motion tracked is effectively keyframed, manually dragging a stroke on a frame that's already been tracked overrides the tracking data at that frame with a reference frame that records the new position of the stroke at that frame. If there's no more tracking data, either before or after that reference frame, then the stroke disappears until you do more tracking. If there's tracking data immediately after a reference frame, then the stroke jumps to the position dictated by that tracking data.
- **Deleting Stroke Tracking at Specific Frames:** To accommodate cases where a subject you're isolating moves behind something and becomes hidden for a range of frames, you have the ability to delete stroke tracking for one or more frames. This lets you eliminate the stroke for however many frames the subject you're tracking is obscured and cannot be sampled. This makes it easy to deal with features that come and go over the duration of a clip, since you can delete stroke tracking over however many ranges of frames the feature disappears. If you've muted frames by mistake, you can either undo or re-track the stroke over those frames.

To delete tracking frames, draw a bounding box on the stroke tracks over the range you want to delete, and then choose Clear Selected Track Data from the Magic Mask option menu.

Option Menu Commands for Removing Strokes and Tracking

The Option menu of the Magic Mask palette presents a number of commands for clearing either strokes or stroke tracking data.

- **Clear All Strokes:** Deletes every stroke in the Stroke list, along with tracking.
- **Clear All Strokes of Current Frame:** Deletes the tracking data at the current position of the playhead on all strokes in the list.
- **Clear All Strokes from In to Out:** If you use the I and O keys to set In and Out points in the Stroke Timeline, this command deletes all tracking data on all strokes within that range.
- **Clear Selected Stroke from In to Out:** If you select a stroke in the Stroke list, and then use the I and O keys to set In and Out points in the Stroke Timeline, this command deletes the tracking data of selected strokes within that range.
- **Clear Selected Track Data:** If you drag a bounding box around tracking data for one or more strokes, this command deletes all tracking data within the box.

An Example Stroke Tracking Workflow

By combining motion tracking, manual reference frames, and stroke muting, you'll be able to make short work of isolating most moving subjects. The following procedure illustrates how you might use these techniques together to make the strokes you've drawn follow along with the features you're isolating.

To track one or more strokes to follow a subject:

- 1 If necessary, move the playhead to the frame where you want to begin isolating the subject you want to mask. For example, if you're isolating a person walking into the room through a door, you probably want to begin your work on a frame where the person is already in the room, and work your way backward and forward from that frame. If your subject is already fully visible at the first frame of the shot, you can start there.
- 2 Draw at least one stroke identifying the subject or feature you want to isolate. You cannot track a stroke to follow the features unless there's at least one stroke.
- 3 Choose whether you want to track all strokes or only selected strokes. By default, all existing strokes with at least one frame in the Stroke Timeline at the position of the playhead will be tracked. However, you can choose to track only selected strokes in the Stroke list by choosing Track Selected Stroke Only in the Magic Mask palette's Option menu.
- 4 Click the Track Forwards to End or Track Backwards to Beginning buttons if you want to motion track the strokes over the remaining duration of the clip.

By default, each stroke is tracked to follow whatever image detail immediately surrounds it; you don't have to define a tracking region, this is done automatically. As tracking proceeds, the tracking bar for each stroke will fill up showing you which frames have been tracked, and which have yet to be tracked. Meanwhile, each stroke being tracked will transform to follow along with simple position and rotation changes in the subject. However, things that obscure the subject, as well as more extreme movements, may cause problems.

Strokes that are tracking a feature that moves off screen are automatically excluded from tracking at the frame where the feature is completely off screen, even if that feature returns onscreen in a later frame.

- 5 (Optional) If a problem occurs during the track, such as the subject being occluded behind something else in the frame (for example, a person's face moves behind a tree as they walk), click the Pause Track button. You can deal with problems in the track in the following ways:
- a) **If a stroke has completely moved off of the subject:** For example, you're tracking a stroke to follow a person and they turn around 180 degrees, so the stroke moves off onto the background. To fix this, pause the track, move the playhead to the first frame where the stroke hasn't followed the subject properly, then use the pointer to drag the stroke onto another part of the subject that's clearly visible. On each frame where you drag the stroke to a new position, a new reference frame will be automatically added. Continue this process until the subject's motion is consistent enough to be trackable again, and you can then resume tracking.
 - b) **If a stroke has moved onto something that's partially obscuring the subject:** Pause the track, drag a bounding box over the frames where the track is having a problem, and choose Delete Selected Tracking from the Magic Mask Option menu to clear the bad tracking data. Then, move the playhead to the first frame of the deleted tracking data, and use the pointer to drag the stroke onto any part of the subject that's visible to either side of the occlusion. Move the playhead to each successive frame, and drag the stroke onto whichever part of the subject is visible from behind the occlusion, until the subject moves past the occlusion. At that point, you can use the Track Forwards to End or Track Backwards to Beginning buttons to continue tracking as usual.
 - c) **If the subject becomes completely hidden:** Clips where the subject appears and disappears over the duration of a clip can be resolved by deleting the stroke or strokes on that subject over all frames where the subject is obscured, so that strokes only sample the image when the subject is visible and thus able to be isolated. You can delete one or more strokes by dragging a bounding box over a range of frames, and then choosing Clear Selected Track Data from the Magic Mask Option menu.
 - d) **If a stroke is having trouble following the subject:** If the subject you're tracking is moving too fast, too irregularly, or is changing shape too greatly for the tracker to give a good result (for example, someone lifts their arm towards the camera), you can track through the problem area of the clip one frame by frame using the Track Forward 1 Frame or Track Backward 1 Frame buttons, and then use the pointer to manually drag the stroke to follow along with the subject as you go. Once the problem motion stops and the motion of the subject becomes more regular, you can use the Track Forwards to End or Track Backwards to Beginning buttons to continue tracking as usual.
 - e) **If the subject is moving off the frame:** If you've drawn a stroke in the middle of someone's face, and their face is moving off screen, you can manually reposition the stroke on the last few frames before the subject completely exits so that stroke continues to identify what parts of the face are still visible even though the part of the face it was originally following becomes hidden. Once the face completely exits the frame, you can mute the stroke for the rest of the shot.
- 6 (Optional) If you only tracked the stroke's movement over part of a shot, you can click the Jump to Start or Jump to End buttons to move the playhead to the first or last tracked frame, in preparation for tracking from that frame onward.